PROCUREMENT PACKAGE:	EL003 - 11kV Switchgear
EQUIPMENT NUMBER(S):	**EQUIPMENT NAME(S):**
F12-SW-012 F16-SW-011 F23-SW-016 F28-SW-013	Primary Crushing 11kV Switchboard Flotation 11kV Switchboard Final Tails Pumps 11kV Switchboard Ultra Fine Grinding 11kV Switchboard
Description:	Air Insulated (AIS) with a bus voltage transformer, bus earth switch, 400A withdrawable fused contactors, 1250A and 630A withdrawable circuit breakers. Bus earth switch shall be key interlocked with incomer circuit breaker to prevent closure with live bus.
Reference Documents	2000-F12-SLD-EL-10001 – F12-SW-012 Primary Crushing 11kV Switchboard Single Line Diagram 2000-F16-SLD-EL-10001 – F16-SW-011 Flotation 11kV Switchboard Single Line Diagram Sheet 1 2000-F16-SLD-EL-10002 – F16-SW-011 Flotation 11kV Switchboard Single Line Diagram Sheet 2 2000-F16-SLD-EL-10003 – F16-SW-011 Flotation 11kV Switchboard Single Line Diagram Sheet 3 2000-F23-SLD-EL-10001 – F23-SW-016 Final Tails Pumps 11kV Switchboard Single Line Diagram Sheet 1 2000-F23-SLD-EL-10002 – F23-SW-016 Final Tails Pumps 11kV Switchboard Single Line Diagram Sheet 2 2000-F28-SLD-EL-10001 – F28-SW-013 Ultra Fine Grinding 11kV Switchboard Single Line Diagram Sheet 1
Reference Specifications	2000-F00-STS-EL-10004 – HV Electrical Testing Specification 2000-F00-STS-EL-10015 – High Voltage Switchboards and Switchgear Specification 2000-F00-STS-GE-10001 – Packing and Shipping Specification 2000-F00-STS-GE-10002 – Site Specific Information 2000-F00-STS-GE-10003 – Plant Numbering Specification

SCOPE OF WORK

Works Included

1. Design, supply, and fabrication of four (4) 11kV switchboards including, but not limited to:

- Withdrawable type switchboard, single tiers
- Air insulated switchboard
- Rating: 1250A bus, 25kA 3 sec, AFLR
- 48VDC auxiliary supply voltage
- Earth switches are to be motorised and equipped with **visual confirmation** - by mirror and light for visual confirmation of earth switch positions
- 1 x bus earth switch electrically interlocked to incomer circuit breakers
- Shunt-trip, Close and No Volt Coils
- Circuit Breaker Remote switching and racking capability (to be implemented by the switchroom builder)
- Fused Contactor Remote switching and racking capability (to be implemented by the switchroom builder)
- Earth switch (cable side) – motorised and to have remote operation capability
- SEL Protection Relays (as per tables below and Single Line Diagrams) with common functions:
    - SEL 751 protection relays with integral arc flash (where available), SEF protection and Modbus TCP and IEC 61850 communications and dual RJ45 ethernet ports
    - SEL 787 Transformer differential protection relay
    - SE-330AU NER monitor relays where specified
    - SEL710 motor protection relays
- SEL735 Power Meters with Modbus TCP communications
- Supplementary SEL 751#JP0G used as arc flash detection relays where SEL relays do not have native arc flash functionality (ie. on 11kV Incomers with SEL787 relays)
- One additional set of Protection CT's for SEL 751#JP0G in each Incomer (for arc flash).
- Common CT specifications:
    - Protection current transformers - 5P20, 2.5 VA with burden to be appropriately sized by the Vendor
    - Metering current transformers - CL 0.2, 2.5 VA Metering current transformers with burden to be appropriately sized by the Vendor
- Common VT specifications: 11/√3 / 0.11/√3 kV 50VA CL0.2 & 50VA CL 3P
- Fortress keys – vendor to liaise with Fortress Distributor for key management

2. All ancillary items and special tools normally supplied with, or required to adjust and maintain all components associated with the equipment

- Circuit breaker trolley for removing Circuit Breakers and Fused contactors
- Switching and/or racking handles;
- Internal arc absorbers - no external arc ducting required. Vendor to provide switchroom pressure calculation.
- Equipment nameplates and labels;
- Bus bar connection pieces;
- Inter-panel wiring;
- Any other items which are normally suppled with this equipment;
- All fasteners, gaskets, packers, shims and any other items necessary for assembly in switchroom.

3. Supply of all necessary documentation, type test certificates (no type test repetition), drawings and test documentation;

4. Routine testing at manufacturer's premises;

5. Provision for on-site support during erection and factory acceptance testing at switchroom vendor premises;

6. Packaging suitable for sea and road transport to Perth-based switchroom vendor premises.

7. Interlocking locks and keys (Fortress), including:

- Incomers - single set (incl. barrel) Fortress and keyed alike to upstream 33kV switchgear. Fortress locks to be managed by Fortress distributor;
- VSD feeders – single key on Fused Contactor earth switch to key exchange box (key exchange box is for VSD power section doors, to ensure cable earth switch is on prior to opening VSD power section doors).

8. F12-SW-012 Primary Crushing 11kV Switchboard (drg. 2000-F12-SLD-EL-10001)

A. Equipment List:

Panel No.	Description	Rating (Amps)	Protection relay	Protection CT ratio	Metering CT ratio	Core Balance CT ratio
1	Bus VT and earth switch	NA	NA	NA	NA	NA
2	Incomer – *Circuit Breaker* - *Additional core of 300/1 A CTs for SEL751#JP0G*	630	1. SEL 787#22EK 2. Littlefuse SE-330AU (NER monitor) 3. SEL 735#L54R	300/1	300/1	NA

3	F12CG401-VS Primary Crusher VSD – *Fused Contactor*	400	SEL 751#JP0G	50/1	NA	50/1
4	- F12-TR-001 LV Transformer feeder – *Circuit Breaker*	630	SEL 751#JP0G	200/1	NA	50/1
5	SPARE – *Circuit Breaker*	630	SEL 751#JP0G	50/1	NA	50/1

9. F16-SW-011 Flotation 11kV Switchboard (drg. 2000-F16-SLD-EL-10001, 10002 and 10003)

A. Equipment List:

Panel No.	Description	Rating (Amps)	Protection relay	Protection CT ratio	Metering CT ratio	Core Balance CT ratio
1	Bus VT and earth switch	NA	NA	NA	NA	NA
2	F16AG421-VS Rougher Flotation Cell 1 Agitator VSD – *Fused Contactor*	400	SEL 751#JP0G	50/1	NA	50/1
3	F16AG422-VS Rougher Flotation Cell 2 Agitator VSD – *Fused Contactor*	400	SEL 751#JP0G	50/1	NA	50/1
4	F16AG423-VS Rougher Flotation Cell 3 Agitator VSD – *Fused Contactor*	400	SEL 751#JP0G	50/1	NA	50/1
5	F16AG424-VS Scavenger Flotation Cell 1 Agitator VSD – *Fused Contactor*	400	SEL 751#JP0G	50/1	NA	50/1
6	F16AG425-VS Scavenger Flotation Cell 2 Agitator VSD – *Fused Contactor*	400	SEL 751#JP0G	50/1	NA	50/1
7	F16AG426-VS Scavenger Flotation Cell 3 Agitator VSD – *Fused Contactor*	400	SEL 751#JP0G	50/1	NA	50/1
8	F16AG427-VS Scavenger Flotation Cell 4 Agitator VSD – *Fused Contactor*	400	SEL 751#JP0G	50/1	NA	50/1
9	Incomer – *Circuit Breaker* - *Additional set of 900/1 A CT's for Vamp 321*	1250	1. SEL 787#22EK 2. Littelfuse SE-330AU (NER monitor) 3. SEL 735#L54R	900/1	900/1	NA
10	F24PW401-VS Mill Water Pump 1 VSD – *Fused Contactor*	40	SEL 751#JP0G	50/1	NA	50/1
11	F24PW402-VS Mill Water Pump 2 VSD – *Fused Contactor*	400	SEL 751#JP0G	50/1	NA	50/1
12	SPARE – *Circuit Breaker*	630	SEL 751#JP0G	50/1	NA	50/1
13/14	Bus Tie – refer SLD – *Circuit Breaker*	1250	NA	NA	NA	NA
15	Generator Incomer – *Circuit Breaker* *(Vendor will supply 3 x 200/1 A CTs plus 1 x 200/1 core balance CT for installation by others into the Generator 11kV terminal box* – EPC *to supply dimensions)*	630	SEL 751#JP0G	200/1	200/1	200/1
16	F24PW417-VS High Pressure Process Water Pump 1 VSD – *Fused Contactor*	400	SEL 751#JP0G	50/1	NA	50/1
17	F24PW418-VS High Pressure Process Water Pump 2 VSD – *Fused Contactor*	400	SEL 751#JP0G	50/1	NA	50/1

18	F17PP455-VS Flotation Tailings Thickener UF Pump 1 VSD – *Fused Contactor*	400	SEL 751#JP0G	50/1	NA	50/1
19	F17PP456-VS Flotation Tailings Thickener UF Pump 2 VSD – *Fused Contactor*	400	SEL 751#JP0G	50/1	NA	50/1
20	F16-TR-008 LV Transformer Feeder – *Circuit Breaker*	630	SEL 751#JP0G	50/1	NA	50/1
21	Bus VT and earth switch	N/A	N/A	N/A	N/A	N/A

10. F23-SW-016 Final Tails Pumps 11kV Switchboard (drg. 2000-F23-SLD-EL-10001 to 10002)

A. Equipment List

Panel No.	Description	Rating (Amps)	Protection relay	Protection CT ratio	Metering CT ratio	Core Balance CT ratio
1	Bus VT and earth switch	NA	NA	NA	NA	NA
2	F23PP466-VS Final Tails Pump 1 VSD – *Fused Contactor*	400	SEL 751#JP0G	50/1	NA	50/1
3	F23PP467-VS Final Tails Pump 2 VSD – *Fused Contactor*	400	SEL 751#JP0G	50/1	NA	50/1
4	Incomer – Circuit Breaker - *Additional core of 300/1 A CTs for SEL 751#JP0G*	630	1. SEL 787#J53Q 2. Littlefuse SE-330AU (NER monitor) 3. SEL 735#L54R	300/1	300/1	NA
5	F23PP468-VS Final Tails Pump 3 VSD – *Fused Contactor*	400	SEL 751#JP0G	50/1	NA	50/1
6	F23PP469-VS Final Tails Pump 4 VSD – *Fused Contactor*	400	SEL 751#JP0G	50/1	NA	50/1
7	F72MB411-VS Lime Slaking Ball Mill VSD – *Fused Contactor*	400	SEL 751#JP0G	50/1	NA	50/1
8	F72-PK-405 Oxygen Plant Feeder 1 - *Circuit Breaker*	630	SEL 751#JP0G	200/1	NA	50/1
9	F72-PK-405 Oxygen Plant Feeder 2 - *Circuit Breaker*	630	SEL 751#JP0G	200/1	NA	50/1
10	SPARE	630	SEL 751#JP0G	50/1	NA	50/1

11. F28-SW-013 Ultra Fine Grinding Switchboard (drg. 2000-F28-SLD-EL-10001)

A. Equipment List

Panel No.	Description	Rating (Amps)	Protection relay	Protection CT ratio	Metering CT ratio	Core Balance CT ratio
1	Bus VT and earth switch	NA	NA	NA	NA	NA
2	F28-MUF-402 Ultra Fine Grinding Mill 2 – *Circuit Breaker + AFD VAMP 321 AFD monitoring as per SLD*	630	SEL 710#55RX	300/1	NA	50/1
3	Incomer – Circuit Breaker - *Additional set of 900/1 A CTs for SEL 751#JP0G*	1250	1. SEL 787#22EK 2. Littlefuse SE-330AU (NER monitor) 3. SEL 735#L54R 4. VAMP 321	900/1	900/1	NA
4	F28-MUF-404 Ultra Fine Grinding Mill 3 – *Circuit Breaker + AFD VAMP 321 AFD monitoring as per SLD*	630	SEL 710#55RX	300/1	NA	50/1
5	SPARE Circuit Breaker	630	SEL 751#JP0G	250/1	NA	50/1

Work Excluded

1. Cabling external to switchboard;
2. Spare parts;
3. Offloading the equipment at vendor premises;
4. Installation and commissioning at site;

DUTY SPECIFICATION	UOM	EPC DATA	SUPPLIER DATA Vendor to complete all data in Supplier Data column any deviations to be noted	NOTES
General				
Switchgear Type	-	Gas (GIS), Air (AIS) and/or Shielded Solid insulated (SIS) or a combination	Air Insulated (AIS)	3
Access required	-	Single sided, front **Manufacturer to state if rear access is required for cable terminations or maintenance eg. VT access**	Single sided, front	1
Internal Arc Fault Containment	-	AFLR	AFLR	1
Partition Class	-	PM	PM	1
Service Continuity Category	-	LSC2A	LSC2A	1
Intended Installation	-	Elevated switchroom, access front only. Vendor to state if rear access is required.	Front access	1
Arc vents/chutes	-	Internal arc quenching (arc absorber)	Top chimneys (No external room venting required)	3
Make	-	Vendor to advise	ABB	2
Model	-	Vendor to advise	ZS1	2
Country of Manufacture	-	Vendor to advise	China	2
Compliance with Australian Standards (or equivalent IEC standards)	Comply/Do not comply	Comply	Yes/IEC62271	1
Drive and Load Descriptions	-	As per Single Line Diagrams	Noted	1
CB ratings	-	As per Single Line Diagrams	Noted	1, 5
Environmental Conditions				
Plant Location	-	Western Australia	Western Australia	1
Service Location	Indoor/Outdoor	Indoor (state any applicable deratings)	Indoor	1
Ambient Air Temperature - Min	°C	0	-5	1
Ambient Air Temperature - Max (inside room)	°C	27	+40	1
Ambient Air Temperature - Max (outside)	°C	50	Noted	1
Humidity	%	0 to 95% for 24h / 90% for 1 month	Noted	1
Altitude	m	<1000m	<1000m	1
Other Data	-	NA		1
Supply				
Number of phases	-	3	3	1
Input Voltage	kV	11 (+/- 10%)	11	1
Frequency	Hz	50 (±2%)	50	1
Neutral Earthing	-	Impedance (limited to 25A) by Neutral Earthing Resistor	Noted	1
Switchboard Ratings				
Rated Voltage	kV	12	12	1
Rated Busbar Current	A	1250	1250	1
Rated Power Frequency Withstand Voltage	kV rms	28	28	1
Rated Lighting Impulse Voltage	kV peak	75	75	1
Rated Short time Withstand Current and Time	kA for sec	25kA / 3 sec, minimum	25kA / 3 sec	3
Rated Internal Arc Fault Containment	kA for sec	21kA / 1 sec, minimum	21kA / 1 sec	3
Rated Peak Withstand Current	kA	62.5kA	62.5kA	3
Endurance Class	Comply/Do Not Comply	E2, M2, C2	Comply	1
Busbar Arrangement	Single / Duplicate	Single	Single	1
Busbar Identification	Yes / No	Yes	Yes	1
Insulated Busbars	Yes / No	Yes	Yes	1
Busbar Material	-	Copper	Copper	2

DUTY SPECIFICATION	UOM	EPC DATA	SUPPLIER DATA Vendor to complete all data in Supplier Data column any deviations to be noted	NOTES
Future Busbar Extension at Either End	Yes / No	Yes (both)	Yes (both)	1
Earth Bar Withstand Current and Time	kA for sec	16kA / 1 sec	Comply	3
Control Voltage	V	48V DC	48V DC	1
Mechanical Interlocks				
Bus Earth Switch / Incoming Circuit Breakers	Comply/Do Not Comply	All incomer circuit breakers Fortress key interlocked with bus earth switch. Refer to single line diagrams.	Comply	1
Incoming Earth Switch / Upstream Circuit Breaker	Comply/Do Not Comply	Incoming earth switches key interlocked with upstream disconnector. Refer to single line diagrams.	Comply	1
Cable Cubicle Access	Comply/Do Not Comply	Cubicle cannot be opened when earth switch is open.	Comply	1
Circuit Breaker / Earth Switch	Comply/Do Not Comply	Earth switch and FC/CB cannot be In Service and Closed at the same time. Refer to single line diagrams.	Comply	1
Visual position confirmation	Comply/Do Not Comply	CB truck position and earth switch to be visible to confirm open and earth switch ON/OFF operation (window or camera)	Comply	1
Fused Contactor Earth switch	Comply/Do Not Comply	Fortress key interlock required between earth switch and VSD compartment	For earthing switch On/Off visual confirmation, Mirror+light is considered, however it is dependent on the power cable terminations, which may hinder the viewing.	1
Incoming Circuit Breakers (Incomers)				
Manufacturer	-	Vendor to advise	ABB	2
Model	-	Vendor to advise	VD4/P	2
Circuit Breaker Type	-	VCB, Withdrawable	VCB Withdrawable	3
Rated Voltage	kV	12kV	12kV	1
Rated Nominal Current	A	1250 & 630	1250 & 630	1
Rated Short time Withstand Current and Time	kA for sec	25 / 3sec, minimum	25kA/3sec	3
Making Current – Peak	kA	Vendor to advise	63	2
Breaking Current – Symmetrical / Asymmetrical	kA	Vendor to advise	25	2
Close Coil Voltage	V	48V DC	48V DC	1
Close Coil Power Demand (energise, latched)	W	Vendor to advise	200W, 5W	2
Tripping Method	-	Shunt Trip and Undervoltage Release	Shunt Trip and Undervoltage Release	1
Shunt Trip Coil Voltage	V	48V DC	48V DC	1
Shunt Trip Coil Power Demand (energise, latched)	W	Vendor to advise	200W, 5W	2
Undervoltage Coil Voltage	V	48V DC	48V DC	1
Undervoltage Coil Power Demand (energise, latched)	W	Vendor to advise	200W, 5W	2
Spring Charging Motor Supply Voltage	V	48V DC	48V DC	1
Spring Charge Power Demand (energise, latched)	W	Vendor to advise	600W, 5W	2
Number of Permissible Circuit Breaker Operations				
at Rated Current	No.	Vendor to advise	E2	2
at Rated Fault Current	No.	Vendor to advise	E2	2
CB Close time	ms	Vendor to advise	50…80	2
CB Open time	ms	Vendor to advise	33…60	2
Number of Auxiliary Contacts	-	Vendor to advise	5NO+5NC	2
Trip circuit supervision relay	Yes / No	Yes, May be integral to protection relay	As per SEL relays' part references/functions offered	1

DUTY SPECIFICATION	UOM	EPC DATA	SUPPLIER DATA Vendor to complete all data in Supplier Data column any deviations to be noted	NOTES
Earth Switch Remote Operation	Yes / No	Yes	No	1
Outgoing Circuit Breakers (Feeders)				
Manufacturer	-	Vendor to advise	ABB	2
Model	-	Vendor to advise	VD4/P	2
Circuit Breaker Type	-	VCB, Withdrawable	VCB, Withdrawable	3
Rated Voltage	kV	12kV	12kV	1
Rated Nominal Current	A	630A	630A	1
Rated Short time Withstand Current and Time	kA for sec	25 / 3sec, minimum	25kA / 3sec	3
Making Current – Peak	kA	Vendor to advise	63	2
Breaking Current – Symmetrical / Asymmetrical	kA	Vendor to advise	25	2
Close Coil Voltage	V	48V DC	48V DC	1
Close Coil Power Demand (energise, latched)	W	Vendor to advise	200W, 5W	2
Tripping Method	-	Shunt Trip and Undervoltage Release	Shunt Trip and Undervoltage Release	1
Shunt Trip Coil Voltage	V	48V DC	48V DC	1
Shunt Trip Coil Power Demand (energise, latched)	W	Vendor to advise	200W, 5W	2
Undervoltage Coil Voltage	V	48V DC	48V DC	1
Undervoltage Coil Power Demand (energise, latched)	W	Vendor to advise	200W, 5W	2
Spring Charging Motor Supply Voltage	V	48V DC	48V DC	1
Spring Charge Power Demand (energise, latched)	W	Vendor to advise	200W, 5W	2
Number of Permissible Circuit Breaker Operations				
at Rated Current	No.	Vendor to advise	E2	2
at Rated Fault Current	No.	Vendor to advise	E2	2
CB Close time	ms	Vendor to advise	50…80	2
CB Open time	ms	Vendor to advise	33…60	2
Number of Auxiliary Contacts	-	Vendor to advise	5NO+5NC	2
Trip circuit supervision relay	Yes / No	Yes, may be integral to protection relay	As per SEL relays' part references/functions offered	1
Earth Switch Remote Operation	Yes / No	Yes	Yes	1
Fused Contactors				
Manufacturer	-	Vendor to advise		2
Model	-	Vendor to advise		2
Fused Contactor Type	-	Withdrawable		3
Rated Voltage	kV	12kV		1
Rated Nominal Current with fuse	A			2
Rated Short time Withstand Current and Time	kA for sec	25 / 3sec, minimum		3
Making Current – Peak	kA	Vendor to advise		2
Breaking Current – Symmetrical / Asymmetrical	kA	Vendor to advise		2
Close Coil Voltage	V	48V DC		1
Fuse contactor burden	W	Vendor to advise		2
Number of operations		Vendor to advise		2
Feeder Protection Relays				
Manufacturer	-	SEL	SEL	3
Model	-	751	SEL-751#JP0G (751002CCC0X70811F90)	3
Supply Voltage	V	48V DC	48V DC	1
Communications Interface	-	Dual RJ45 Ethernet	As per SEL-751#JP0G	3
Communications Protocols Supported	-	Modbus TCP		3
Protection Relay Features				
Voltage Sensing	Yes / No	Yes	As per SEL-751#JP0G	1
Tripping Supply Monitoring	Yes / No	Yes		1
CT Saturation Detection	Yes / No	No		2
Harmonic Blocking (2nd & 5th)	Yes / No	Yes		1

DUTY SPECIFICATION	UOM	EPC DATA	SUPPLIER DATA Vendor to complete all data in Supplier Data column any deviations to be noted	NOTES
Directional Protection	Yes / No	No		1
Undervoltage Protection	Yes / No	Yes		1
Underfrequency Protection	Yes / No	Yes, on Outgoing Protection Relays only		1
Transformer Protection Relays				
Manufacturer	-	SEL	SEL	3
Model	-	787	SEL-787#22EK (07872EE2CCC0X78810690)	3
Supply Voltage	V	48V DC	48V DC	1
Communications Interface	-	Dual RJ45 Ethernet	As per SEL-787#22EK Transformer Protection Relay	3
Communications Protocols Supported	-	Modbus TCP		3
Protection Relay Features (essential)				
2 Winding Differential	Yes / No	Yes	As per SEL-787#22EK Transformer Protection Relay	1
Tripping Supply Monitoring	Yes / No	Yes		1
CT Saturation Detection	Yes / No	No		2
Harmonic Blocking (2nd & 5th)	Yes / No	Yes		1
LV Restricted Earth Fault	Yes / No	Yes		1
Transformer Alarms &Trips				
Buchholz	Yes / No	Yes	As per SEL-787#22EK Transformer Protection Relay	1
Winding Temp.	Yes / No	Yes		1
Fluid Temp.	Yes / No	Yes		1
Pressure Relief	Yes / No	Yes		1
Power Quality Meters				
Manufacturer	-	SEL	SEL	3
Model	-	735	SEL-735#L54R	3
Supply Voltage	V	48V DC	48V DC	1
Communications Interface	-	Ethernet RJ45	As per SEL-751#JP0G	3
Communications Protocols Supported	-	Modbus TCP		3
Generator Protection				
Manufacturer	-	SEL	SEL	3
Model	-	751	SEL-751#JP0G (751002CCC0X70811F90)	3
Supply Voltage	V	48V DC	48V DC	1
Communications Interface	-	Dual RJ45 Ethernet	As per SEL-751#JP0G	3
Communications Protocols Supported	-	Modbus TCP		3
Motor Protection (LRS)				
Manufacturer	-	SEL	SEL	3
Model	-	710	SEL-710#55RX	3
Supply Voltage	V	48V DC	48V DC	1
Communications Interface	-	Dual RJ45 Ethernet	As per SEL-710#55RX Motor Protection Relay	3
Communications Protocols Supported	-	Modbus TCP		3
Number of RTD inputs (min.)		8		3
Current Transformers				
Phase Current Transformers				
Manufacturer & Model	-	Vendor to advise	ABB	2
Type	Block / Toroid	Vendor to advise	Block	2
Duty	-	Overcurrent Protection & Metering (incomers) Overcurrent and SEF Protection (feeders)	Comply	1, 6
Ratios	-	Refer to SLDs All outgoing feeders have 50/1 core balance CTs	As per offer technical specs., 50/1 core balance CTs are included	3, 6
Class	-	Protection 5P20 2.5VA Metering CL 0.2 2.5VA	Comply	3, 6
Voltage Transformer				
Manufacturer	-	Vendor to advise	ABB	2
Type	-	Vendor to advise	TBA	2

DUTY SPECIFICATION	UOM	EPC DATA	SUPPLIER DATA Vendor to complete all data in Supplier Data column any deviations to be noted	NOTES
Ratio	-	11kV/√3 / 110V/√3	11kV/√3 / 110V/√3	1
Rating	-	50VA & 50VA	15VA & 20VA	3
Class	-	CL0.2 & CL 3P	Comply	3
Fixed or Withdrawable	-	Fixed	Fixed for Riser panel, Withdrawable for Bus-VT panels	1
Cubicle Details				
Ingress Protection Rating	-	IP3X	IP4X	3
Cubicle Control Voltage	V	48V DC	48V DC	1
Local Indication LED Lamps				
Supply Voltage	V	48V DC	48V DC	1
Type	-	Semaphore Type LED Indicator Lamps	Comply	1
Required Indication	-	Circuit Breaker Status Earth Switch Status Interlock Status	Comply, with hard-wired electrical interlocks	1
3 Phase Cable Side Capacitive Phase Indicator	Yes / No	Yes	Yes	1
Anticondensation Heater Supply	V	240V AC	240V AC	3
Anticondensation Heater Power	W	Vendor to advise	100W	2
Electric Controls				
Local/Remote Selector	Yes / No	Yes	Yes	1
Local Trip	Yes / No	Yes	Yes	1
Local Close	Yes / No	Yes	Yes	1
Remote Control Interface (Hardwired to airlock)				
Remote Control Selected	Yes / No	Yes – Dry Contact	Comply	1
Earth Switch Open/Closed	Yes / No	Yes – Dry Contact (Changeover)	Comply	1
Circuit Breaker Open/Closed	Yes / No	Yes – Dry Contact (Changeover)	Comply	1
Interlock Status	Yes / No	Yes - Dry Contact	Comply as per detailed engineering design	1
Circuit Breaker Open	Yes / No	Yes – Remote 48V Signal		1
Circuit Breaker Close	Yes / No	Yes – Remote 48V Signal		1
Circuit Breaker/Fused Contractor truck position	Yes/No	Yes – racked in and racked out		
External Paint Colour	-	Vendor to advise	RAL7035	2
Internal Paint Colour	-	Vendor to advise	Unpainted High quality glossy finish (ALUZINC)	2
Paint Specification	-	Vendor to advise	As attached	2
Viewing Windows/Ports/Cameras				
Type of device to enable visual confirmation of earth switch	Window/Camera	Vendor to advise type	Window with mirror & light (No camera)	2
Window/Camera to view Bus Earth Switch On/Off	Yes / No	Yes		3
Window to view Cubicle Earth Switch On/Off	Yes / No	Yes		3
Light to Assist Viewing Earth Switch	Yes / No	Yes		3
Maintenance Access				
Access to VT	Front / Rear / Top	Vendor to advise	Front	2
Access to Busbar Joints	Front / Rear / Top	Vendor to advise	Top	2
Access to Maintenance Connections	Front / Rear / Top	Vendor to advise	Front	2
Access to Cable Connections	Front / Rear	Vendor to advise	Front	2
Cable Connections				
Cable Entry	Top / Bottom	Bottom	Bottom	1
Cable Connection Type	Type C/Flags	Vendor to advise	Pls. refer to ZS1 IOM	1
Incoming Cable Type	-	XLPE	Noted	1
Maximum Cable Size	mm^2	4 x 3C 120mm^2	Comply	1
Outgoing Cable Type	-	XLPE	Noted	1
Maximum Cable Size	mm^2	1 x 3C 150mm^2	Comply	1
Recommended Termination Kit	-	Vendor to advise	TBA	2
Gland Plate Window Size (L x W)	mm x mm	Vendor to advise	Pls. refer attached drawings	2
Height from Gland Plate to Cable Terminals	mm	Vendor to advise	~500mm approx.	2

DUTY SPECIFICATION	UOM	EPC DATA	SUPPLIER DATA Vendor to complete all data in Supplier Data column any deviations to be noted	NOTES
Phase and Wire Colours				
Phase Colours	Comply/Do Not Comply	Red, White, Blue	Comply	1
Neutral Colour	Comply/Do Not Comply	Black	Comply	1
Earth Colour	Comply/Do Not Comply	Green / Yellow	Comply	1
48V DC Tripping and closing (+ve)	Comply/Do Not Comply	Grey	Comply	1
48V DC Tripping and closing (-ve)	Comply/Do Not Comply	Black with white tracing	Comply	1
Labelling Required				
Switchboard Label	-	Etched traffolyte label, centrally located, with Switchboard Equipment Number, Description and SLD number at minimum.	ABB standard Gravoply labels are considered.	1
Switchboard/Tier Nameplate Label	-	Etched traffolyte switchboard nameplate label centrally located. Each tier shall have a nameplate label relevant to the tier contents.		1
Tier/Cubicle Label	-	Etched traffolyte Label per tier, with Tier / Cubicle Number, service description and fed from location (on incoming cubicle) at minimum.		1
Interlock Details	-	Etched traffolyte label per interlock, detailing interlock type and procedure to place interlock in service / out of service.		1
Other Cubicle Labelling	-	Etched traffolyte label per item (e.g. protection relay, indicator lamp, switches) detailing applicable equipment number and description.		1
Earth Switch and Circuit Breaker Isolation Points		Etched traffolyte label at each isolation point.		1
Test Requirements				
Type	-	AS62271-200 Sub Clause 6.1 (or equivalent)	Pls. refer attached type test certification	3, 4
Routine	-	AS62271-200 Clause 7 (or equivalent)	Manufacturer standard routine tests based on IEC 62271-200	3
Special	-	N/A		1
Shipping and Delivery				
Packaging	-	Vendor to advise - Suitable for sea	Suitable for sea freight	2
Shipping Section Dimensions	(L x W x H)	Vendor to advise	TBA	2
Weight of Individual Panel/Tier	kg	Vendor to advise	As mentioned in offer	2
Overall Dimensions of switchboard assembly	(L x W x H)	Vendor to advise	As mentioned in offer	2
Number of Shipping Sections	-	Vendor to advise	TBA	2

Notes:

1. Data supplied by EPC The Supplier shall comply with this data, as a minimum. Supplier shall state their applicable ratings/data.
2. Data shall be provided by Supplier.
3. Data proposed by EPC The Supplier shall confirm data or state alternative.
4. Type test certificates to be provided with tender documentation.
5. Tender documentation shall include detailed single line diagram which is reflective of the supplier's offer.
6. The supplier shall advise whether multiple CTs or single CT with two secondary windings shall be utilised for incomer modules.